HANAN HAMED ALHARBI

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EDUCATION

King Abdullah University of Science and Technology (KAUST), *MS in Computer Science* | Thuwal, Saudi Arabia — August 2024

Courses: Deep Learning for Visual Computing | Machine Learning | Deep Generative Modeling | Computational Imaging | Computer Graphics | Digital Design

Umm Al Qura University, *BS in Computer Science* | Makkah, Saudi Arabia — May 2020

Awards: Hail University National Hackathon, **2nd** place, Best Bachelor's Senior Project, **3rd** place & Best Computer Graphics Project, **1st** place

EXPERIENCE

Elm Company, *Senior data Analyst* (Riyadh, Saudi Arabia) — Mar 2025 – Present

- Collaborating with the Research and AI department in the Computer Vision Lab to design and implement real-world applications.
- Specializing in **Geospatial AI** and remote sensing, and contributing to the development of **foundation models** for large-scale geospatial and computer vision tasks.
- Contributing to designing and delivering proofs of concept (POCs) and prototypes to validate research, accelerate deployment, and showcase real-world applications.

Image and Video Understanding Lab (IVUL) - KAUST, *Master's student* (Thuwal, Saudi Arabia) — Sep 2023 – Aug 2024

- Conducted research focused on self-supervised representation learning for both images and videos and learning its techniques for improving performance in downstream tasks such as segmentation and action recognition.
- **Thesis:** *Motion-Aware Siamese Masked Autoencoders for Efficient Video Representation Learning* — Developed a novel approach leveraging a Siamese architecture to incorporate motion information with video masked autoencoder, ensuring improved effectiveness in representation learning.

King Abdullah University of Science and Technology (KAUST), *Teaching Assistant* (Thuwal, Saudi Arabia) — Jan 2024 – May 2024

- **Course:** Deep Learning for Visual Computing. I contributed to preparing course assignments and projects, grading, and offering personalized support to students to enhance their understanding and academic performance.

YNMO, *Front-End Developer* | Full-time (Makkah, Saudi Arabia) — Jan 2021 – Jun 2022

- Developed key functionalities for the web application and contributed to identifying and resolving software bugs.
- Proposed and initiated the development of a recommendation system, focusing on data preparation, processing, and the implementation of the underlying model to enhance the application's user experience.
- Tools: HTML, CSS, JavaScript, TypeScript, Angular, AngularJS, Python, Jupyter notebook, SQL.

WadiMakkah, *Machine Learning Developer - Internship* | Full-time (Makkah, Saudi Arabia) — Jun 2019 – Aug 2019

- Focused on mastering computer vision techniques, executing several mini-projects such as facial recognition.
- Developed and fine-tuned a model for detecting driver distraction, successfully deploying it onto a Raspberry Pi for instantaneous analysis in a capstone project.
- Tools: Python, Raspberry Pi, Anaconda.

SKILLS

Languages	Python, C/C++, Java, JavaScript, TypeScript, Matlab, Bash, LaTeX, Verilog (HDL)
Frameworks	Pytorch, Scikit-Learn, Angular, AngularJS
Software and Tools	Linux, Git, Anaconda
Certifications	Data Science Bootcamp, SDAIA (2021) | Artificial Intelligence, Samsung Innovation Campus (2020) | Deep Learning, Udacity Nanodegree (2019) | AI Programming with Python, Udacity Nanodegree (2018) | Front-End Web Developer, Udacity Nanodegree (2018)
Soft Skills	Leadership, Problem-solving, Documentation, Engaging Presentation.

RELEVANT EXPERIENCES AND PROJECTS

Road Damage Management System | Senior Project — May 2020

- Led a team in the execution of a project aimed at detecting and reporting road distresses in Makkah city, utilizing YOLOv3 technology, integrated with a full system of web and mobile applications.
- My contributions were focused on the development of the AI model and the creation of an accompanying Android application.

Detection and Classification of Driver Behavior | Internship Capstone Project Aug 2019

- Executed the development and deployment of a Convolutional Neural Network (CNN) model on a Raspberry Pi, integrated with a camera and an alert system. This setup was specifically designed to identify distractions during driving, enhancing vehicular safety through real-time monitoring and notifications

Topic Modeling on Amazon Customers' Reviews | Samsung Innovation Campus Capstone Project Sep 2020

- Developed a tool for annotating customer reviews with relevant topics, employing BERT, LDA, and Autoencoder techniques. This tool helps users in filtering reviews according to their interests.